Mirco **Colosi, Ph.D.**

Robotics & Computer Vision Engineer

About me

Passionate about tech, coding and challenges. Love to travel, DIY, and videogames.

Personal

Mirco Colosi
10.10.1991
nationality: Italian
based in Zurich, CH
mirco.colosi@gmail.com

Areas of specialization

SLAM • Computer Vision
State Estimation • Calibration
Sensor Fusion

Skills & Tools

Robotics / Classical & Gen AI
ROS2 / Pytorch / Tensorflow
CI / Unit Testing / Docker

Highlights

Experience with AI
Agile and Scrum
Detail and efficiency-driven attitude
Analytical thinking
Proactive and results-oriented

Where can you find me

mirco-colosi
mircolosi
Mirco Colosi

Short Resumé

2026–present

Senior Vision & Perception Engineer
Neura Robotics · Zurich

- **Custom Middleware support:** Development of custom middleware functionalities for real-time robotic applications.
- **Sensor driver integration:** Integration of various sensors drivers in custom middleware.
- **Humanoid State Estimation:** Implementation of InvEKF algorithm to address humanoid state estimation for robot pose estimation.
- **Navigation Stack for Humanoid Robot:** Implementation of the robot navigation stack that uses the robot estimated state to navigate in the environment.

2021–2026

Research Scientist
Robert Bosch GmbH · Robert Bosch Campus – Renningen

- **Neural simluation & reconstruction:** Implementation of 3D Gaussian Splatting and Gaussian RayTracing algorithms to achieve closed-loop simulations from real-world data.
- **Visual SLAM:** Implementation of vSLAM frontend and integration to existing multi-sensor SLAM pipeline.
- **Perception:** Development of algorithms to extract features and primitives from raw sensor data; Classical and AI-based CV methods for image feature extraction.
- **Cross-sensor calibration:** Development of calibration procedure for robot equipped with heterogeneous sensors.
- **Mobile robot perception, navigation, and behavior:** Development of high-level controller for mobile robot in close-to-obstacle situations; 2D LiDAR sensor data processing for obstacle avoidance; High-level planning using Behavior Trees.
- **Teams management** for delivering new features in large robotic project.
- **Part of some internal task forces** for evaluation of external partners and quick develop of novel software components.
- **Contribution on fleet management system software:** Development of VDA5050-based UGV simulation and visualization.
- **Ph.D. Candidate Mentoring:** Supervision of Ph.D. candidate with successful publication in 1st-Tier conferences and open source code distribution.
- **Open source contribution to ROS2 repositories.**

2017–2021

Ph.D. in Engineering in Computer Science. Focus topic: SLAM
· Rome

Standardizing SLAM: exploiting recurrent patterns for modularity and behavioral robustness. Developed a novel standardized architecture for multi-sensor SLAM systems, and a behavioral controller to enhance robustness based on real-time environmental unforeseen events.

Active part of research community: invited talks

Education

2021

Ph.D. in Engineering in CS
Grad: Excellent
Sapienza University of Rome
Sponsor Robert Bosch GmbH

2017

M.Sc. in AI and Robotics
GPA: 110/110
Sapienza University of Rome

2014

B.Sc. in Computer Engineering
GPA: 102/110
University of Catania

Programming Languages

C++	● ● ● ● ●
Python	● ● ● ● ○
Java	● ● ○ ○ ○

Languages

Italian	mother tongue
English	C1 ● ● ● ● ●
German	A2 ● ● ○ ○ ○

Publications (relevant list)

2024 **RelationField: Relate Anything in Radiance Fields**
arXiv (pre-publish CVPR 2025) · Nashville TN, USA
Sebastian Koch, Johanna Wald, **Mirco Colosi**, Narunas Vaskevicius, Pedro Hermosilla, Federico Tombari, Timo Ropinski

2024 **Open3DSG: Open-vocabulary 3d scene graphs from point clouds with queryable objects and open-set relationships**
CVPR · Seattle WA, USA
Sebastian Koch, Narunas Vaskevicius, **Mirco Colosi**, Pedro Hermosilla, Timo Ropinski

2020 **Plug-and-Play SLAM: A Unified SLAM Architecture for Modularity and Ease of Use**
IROS · Las Vegas, Nevada, USA
Mirco Colosi, Irvin Aloise, Tiziano Guadagnino, Schlegel Dominik, Bartolomeo Della Corte, Arras Kai O., Giorgio Grisetti

2019 **Better lost in transition than lost in space: Slam state machine**
IROS · Macau, China
Mirco Colosi, Sebastian Haug, Peter Biber, Kai O Arras, Giorgio Grisetti

2018 **Proslam: Graph slam from a programmer's perspective**
IROS · Brisbane, Australia
Dominik Schlegel, **Mirco Colosi**, Giorgio Grisetti

Coursera Certificates

Feb. 25, 2025	IBM AI Engineering by IBM
July 23, 2024	DeepLearning.AI TensorFlow Developer by DeepLearning.AI
Jun. 17, 2024	TensorFlow: Advanced Techniques by DeepLearning.AI
Jan. 5, 2024	Deep Learning by DeepLearning.AI
Sep. 29, 2023	Machine Learning by Stanford University & DeepLearning.AI